

A Three-Dimensional VOF Phase-Change Simulation Framework

for a Subcooled Pool-Boiling Chamber with a Submerged Tube-Bundle Condenser

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Abstract

This paper specifies a three-dimensional volume-of-fluid (VOF) phase-change simulation framework for a closed, subcooled pool-boiling chamber with a partially submerged tube-bundle condenser, built on the true CAD geometry of the apparatus. It documents a verified body-fitted mesh (299 656 cells, passing `checkMesh`), the governing equations and numerical scheme for the incompressible two-phase mixture with interface capturing, buoyancy, surface tension, and a Lee phase-change closure, the condenser coolant-side flow routing, and the measured boiling curves and thermal resistances that serve as validation targets. A calibration-to-validation workflow is defined in which the phase-change coefficients are fit against a measured boiling curve and critical heat flux and the converged solution is then compared point by point against experiment. A reduced-cost two-dimensional demonstration establishes that the pipeline runs and behaves physically while showing why quantitative calibration requires the resolved three-dimensional mesh. Consistent with the project’s integrity standard, the framework is presented as a specification with verification references only: no unconverged or surrogate field is reported as a phase-change result, and the calibration and overlay against experiment await the converged production run.

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1 Overview and scope of the CFD framework

The apparatus modeled here is the closed subcooled pool-boiling chamber with a partially submerged tube-bundle condenser of Shukla and Kandlikar [1], whose thermal performance across two evaporator surfaces, two condenser bundles, and four coolant inlet temperatures is characterized and predicted in a companion reduced-order study of the same apparatus; the present work develops the complementary high-fidelity route, a three-dimensional VOF simulation with phase change that resolves the boiling and condensation fields on the real geometry.

This document specifies that framework completely and reproducibly, and it assembles the experimental reference, coolant operating conditions, measured boiling curves, and thermal resistances for both condensers, that the simulation is required to reproduce. Following the integrity standard of the broader project, we keep a hard line between three sources: measured data, the validated reduced-order model, and the CFD. The boiling-curve and resistance comparisons in Section 7 are *experiment versus the reduced-order model*; they are the CFD’s target, not CFD output. The CFD field contours and the CFD-to-experiment overlay require the converged run of Section 9 and are not fabricated here.

2 Apparatus geometry

The full apparatus geometry, including both evaporator surfaces and both condenser bundles, is documented in the companion reduced-order study; this section records only the geometric detail needed to reproduce the CFD mesh.

2.1 Chamber and computational domain

The domain is the chamber interior, $80 \times 80 \times 42$ mm (x width, y tube axis, z height), gravity along $-z$ (Fig. 1). The 42 condenser tubes are imported as a triangulated surface at their measured CAD positions; the bundle straddles the liquid level so the lower rows are submerged and the upper rows sit in the vapor space. The heated chip (34.5×32 mm, 11.04 cm^2 footprint) is centered on the floor, and the pool fills the lower 16.8 mm (about 40%). Figure 2 gives the dimensioned cross-section with the real staggered tube layout.

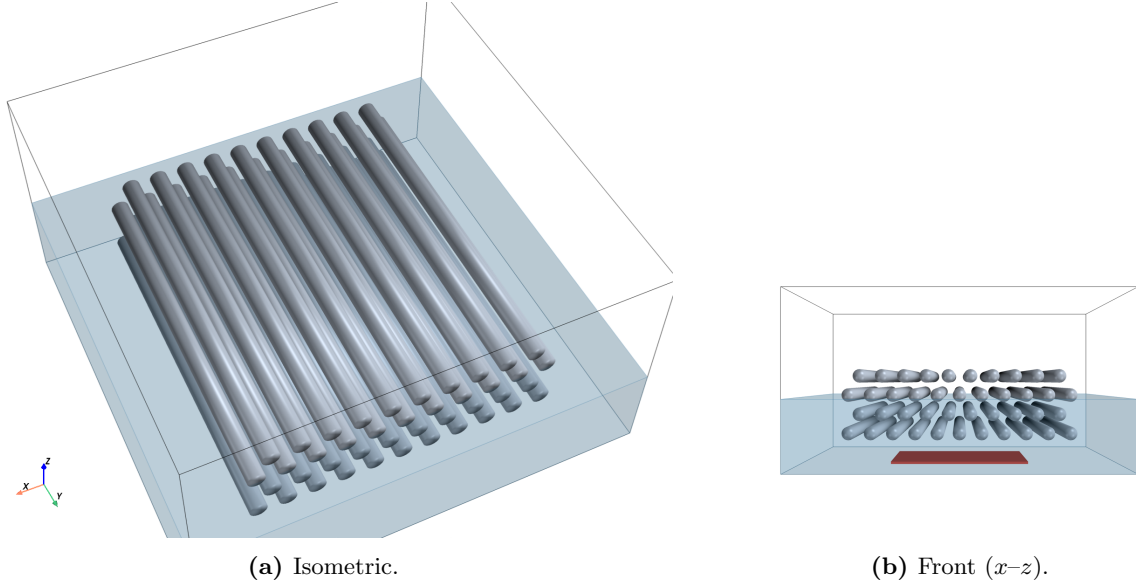


Figure 1: Computational domain: chamber (outline), 42-tube bundle (grey), heated chip (red), subcooled pool (blue) at 40% fill. The bundle straddles the interface.

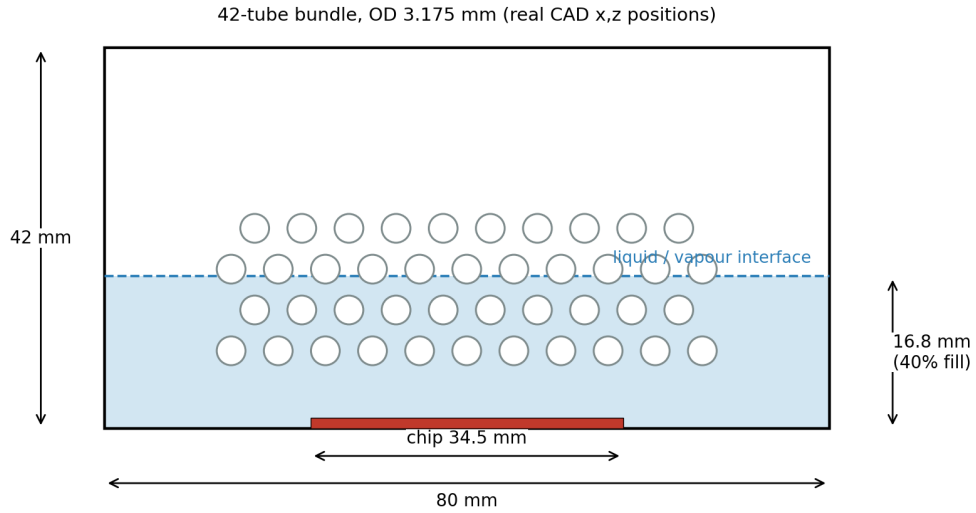
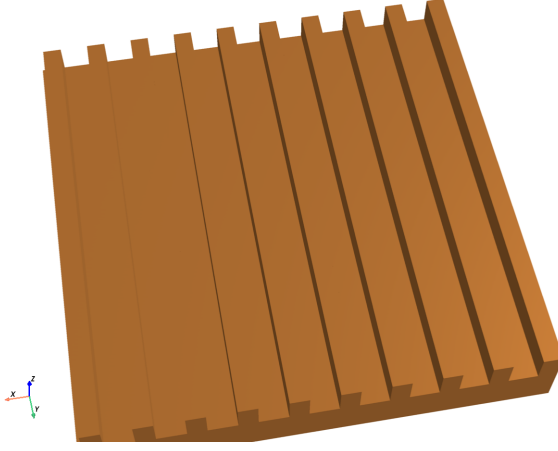


Figure 2: Dimensioned x - z cross-section with the measured 42-tube layout; the dashed line is the initial interface at 16.8 mm.

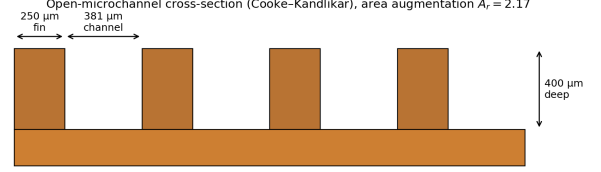
2.2 Evaporator surfaces

Both chips share the 11.04 cm^2 copper footprint; the plain chip is a smooth boiling surface and the enhanced chip carries the open Cooke–Kandlikar microchannels detailed in the companion reduced-order study (Fig. 3). In the CFD the plain surface is modeled as a flat heated wall; resolving or sub-grid-modeling the microchannel surface is a planned extension (Section 11).

open-microchannel surface (6 mm portion)



(a) Surface portion (6 mm).



(b) Cross-section with dimensions.

Figure 3: Open-microchannel evaporator surface (Cooke–Kandlikar geometry).

2.3 Condenser bundles

Two bundles are studied (Fig. 4, Table 1). The 33-tube bundle uses fewer, larger tubes (OD 4.76 mm) with open manifolds; the 42-tube bundle uses more, smaller tubes (OD 3.175 mm) in a confined in-chamber manifold. The CFD case documented here is the 42-tube geometry; a 33-tube case is a separate build (Section 11).



33-tube (OD 4.76 mm, schematic layout)

42-tube (OD 3.175 mm, CAD)

Figure 4: The two condenser bundles to scale: 33-tube (OD 4.76 mm; layout schematic, CAD positions not available) and 42-tube (OD 3.175 mm, from CAD).

Table 1: Condenser bundle geometry.

Bundle	Tubes	OD (mm)	ID (mm)	L (mm)	Layout
33-tube	33	4.76	3.14	95.0	open manifold
42-tube	42	3.175	1.39	62.6	confined

3 Governing equations

The solver `interCondensatingEvaporatingFoam`, part of the OpenFOAM finite-volume framework [2], advances a single incompressible two-phase (water/vapor) mixture with volume-of-fluid interface capturing [3], buoyancy, surface tension, and a volumetric phase-change source. The flow is treated as laminar (justified in Section 3.5). A single momentum and energy field is solved for the mixture, with the two phases distinguished by the liquid volume fraction $\alpha \equiv \text{alpha.water} \in [0, 1]$.

3.1 Two-phase VOF transport

Mass conservation of each phase, with interconversion $\dot{m}$ (mass per unit volume per unit time, positive for evaporation), gives the indicator-function transport equation solved with an interface-compression flux $\mathbf{U}_r$ that counteracts numerical smearing,

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\mathbf{U}\alpha) + \nabla \cdot (\mathbf{U}_r \alpha(1 - \alpha)) = \frac{\dot{m}_c - \dot{m}_e}{\rho_l}, \quad \mathbf{U}_r = c_\alpha |\mathbf{U}| \frac{\nabla \alpha}{|\nabla \alpha|}, \quad (1)$$

where $c_\alpha = 1$ sets the compression strength. The compression term acts only in the interface band ($0 < \alpha < 1$) and vanishes in the bulk. Because mass is exchanged between phases of unequal density, the mixture velocity is not divergence-free; continuity becomes

$$\nabla \cdot \mathbf{U} = \dot{m} \left(\frac{1}{\rho_v} - \frac{1}{\rho_l} \right), \quad (2)$$

so evaporation is a local volume source and condensation a local sink. Mixture properties are volume-fraction weighted,

$$\rho = \alpha \rho_l + (1 - \alpha) \rho_v, \quad \mu = \alpha \mu_l + (1 - \alpha) \mu_v, \quad (\rho c_p) = \alpha \rho_l c_{p,l} + (1 - \alpha) \rho_v c_{p,v}. \quad (3)$$

3.2 Momentum and surface tension

Momentum is solved for the mixture in the buoyant p_{rgh} form, where $p_{rgh} = p - \rho \mathbf{g} \cdot \mathbf{x}$ removes the hydrostatic head from the pressure solution and improves conditioning,

$$\frac{\partial(\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U} \mathbf{U}) = -\nabla p_{rgh} - (\mathbf{g} \cdot \mathbf{x}) \nabla \rho + \nabla \cdot [\mu (\nabla \mathbf{U} + \nabla \mathbf{U}^T)] + \mathbf{F}_{st}. \quad (4)$$

The second term is the buoyancy force carried by density gradients (the engine of natural convection in the subcooled pool). Surface tension enters as a continuum surface force [4] localized at the interface,

$$\mathbf{F}_{st} = \sigma \kappa \nabla \alpha, \quad \kappa = -\nabla \cdot \left(\frac{\nabla \alpha}{|\nabla \alpha|} \right), \quad (5)$$

with κ the interface curvature. Velocity is coupled to pressure through Eq. (2) in a PISO loop (Section 6).

3.3 Energy and the latent coupling

Energy is solved in temperature form for the mixture, with conduction through an effective conductivity and a volumetric latent sink/source at the interface,

$$\frac{\partial(\rho c_p T)}{\partial t} + \nabla \cdot (\rho c_p \mathbf{U} T) = \nabla \cdot (k_{\text{eff}} \nabla T) - \dot{m} h_{fg}, \quad k_{\text{eff}} = \alpha k_l + (1 - \alpha) k_v. \quad (6)$$

Evaporation removes $\dot{m}_e h_{fg}$ from the liquid at the heated surface; condensation releases $\dot{m}_c h_{fg}$ at the cold tubes. The same effective conductivity defines the chip heat-flux probe used to build the simulated boiling curve (Section 9).

3.4 Phase-change closure

The interfacial mass flux is closed with the constant-coefficient (Lee) model [5], in which the rate is proportional to the local departure from saturation, driving the metastable cells back toward T_{sat} (Fig. 5),

$$\dot{m}_e = C_E \alpha \rho_l \frac{T - T_{\text{sat}}}{T_{\text{sat}}} \quad (T > T_{\text{sat}}), \quad \dot{m}_c = C_C (1 - \alpha) \rho_v \frac{T_{\text{sat}} - T}{T_{\text{sat}}} \quad (T < T_{\text{sat}}), \quad (7)$$

where C_E (**coeffE**) and C_C (**coeffC**) are relaxation frequencies (s^{-1}). The coefficient sets an interfacial relaxation time $\tau \sim 1/(C \Delta T/T_{\text{sat}})$: too small and the interface superheats unphysically; too large and the equations stiffen and the time step collapses. There is no universal value, C_E and C_C are mesh- and resolution-dependent and are calibrated against the measured boiling curve (Section 7); this is the central empirical content of the model and the reason a calibration campaign, not a single run, is required.

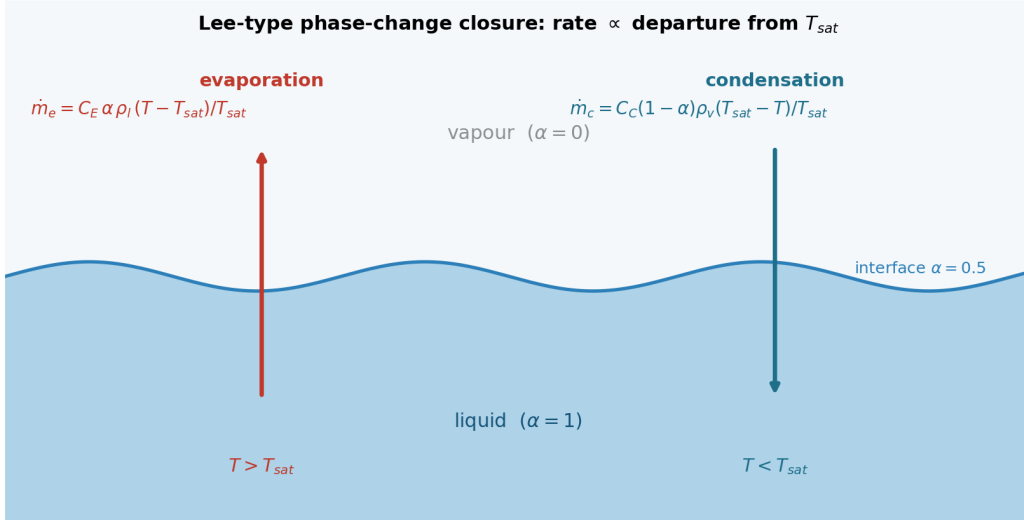


Figure 5: Phase-change closure: evaporation where the cell is superheated ($T > T_{\text{sat}}$) and condensation where it is subcooled ($T < T_{\text{sat}}$), each at a rate proportional to the departure from saturation. The interface is the $\alpha = 0.5$ surface.

3.5 Flow regime and dimensionless groups

Table 2 collects the governing dimensionless groups for water at the operating point. Three features shape the modeling choices. First, the pool Rayleigh number ($Ra \sim 9 \times 10^6$, based on the liquid depth and the chip-to-pool difference) sits well below the $\sim 10^9$ transition for a heated horizontal layer, so the bulk buoyant flow is laminar; the vigorous agitation in the chamber is driven by boiling and bubble departure rather than by turbulent natural convection, which supports the laminar mixture treatment with the interface resolved. Second, the Bond and confinement numbers based on tube diameter ($Bo = 1.5\text{--}3.4$, $Co = L_c/D = 0.5\text{--}0.8$) show the capillary length $L_c = \sqrt{\sigma/\rho_l g} \approx 2.6\text{ mm}$ is comparable to the tube spacing, so capillarity and confinement matter at the bundle, the smaller-bore confined 42-tube bundle is more strongly affected than the open 33-tube. Third, the liquid-to-vapor density ratio is large ($\rho_l/\rho_v \approx 3300$) and the superheat Jakob number is small ($Ja = c_{p,l}\Delta T_{sup}/h_{fg} \approx 0.04$); the large density ratio is the main source of numerical stiffness (the volume source in Eq. (2) and the property jump), which is why the time step is interface-Courant limited. The chamber operates in nucleate boiling up to CHF (Fig. 6); the model is not intended for the transition or film-boiling branches.

Table 2: Dimensionless groups (water, operating point: $\Delta T_{sup} \approx 20\text{ K}$, $\Delta T_{sub} \approx 15\text{ K}$, liquid depth 16.8 mm).

Group	Definition	Value	Role
Capillary length	$L_c = \sqrt{\sigma/\rho_l g}$	2.6 mm	bubble / meniscus scale
Prandtl (liquid)	$\mu c_p/k$	2.7	momentum vs thermal diffusion
Jakob (superheat)	$c_{p,l}\Delta T_{sup}/h_{fg}$	0.036	sensible vs latent at the chip
Jakob (subcooling)	$c_{p,l}\Delta T_{sub}/h_{fg}$	0.027	condensation driving
Density-ratio Jakob	$\rho_l c_{p,l}\Delta T_{sup}/(\rho_v h_{fg})$	117	vapour generation intensity
Bond (42 / 33-tube)	$\rho_l g D^2/\sigma$	1.5 / 3.4	gravity vs capillarity at a tube
Confinement (42 / 33)	L_c/D	0.82 / 0.55	bundle confinement
Rayleigh (pool)	$g\beta\Delta T H^3/\nu\alpha_{th}$	9×10^6	laminar bulk convection
Density ratio	ρ_l/ρ_v	3300	numerical stiffness

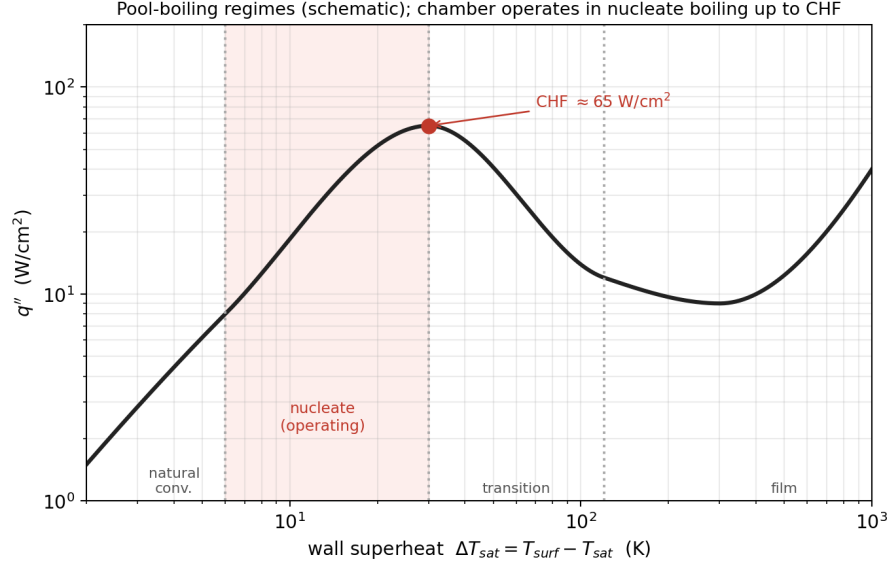
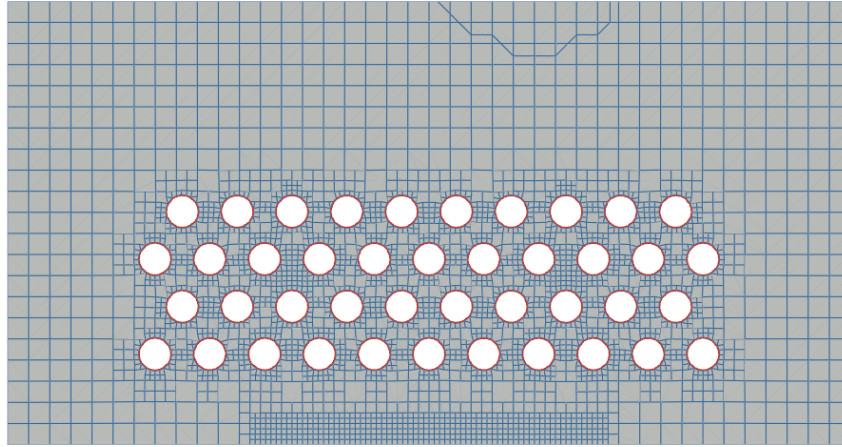


Figure 6: Pool-boiling regime map (schematic). The chamber operates on the nucleate-boiling branch up to the measured CHF; the constant-coefficient closure is calibrated for this branch, not for transition or film boiling.

4 Mesh generation

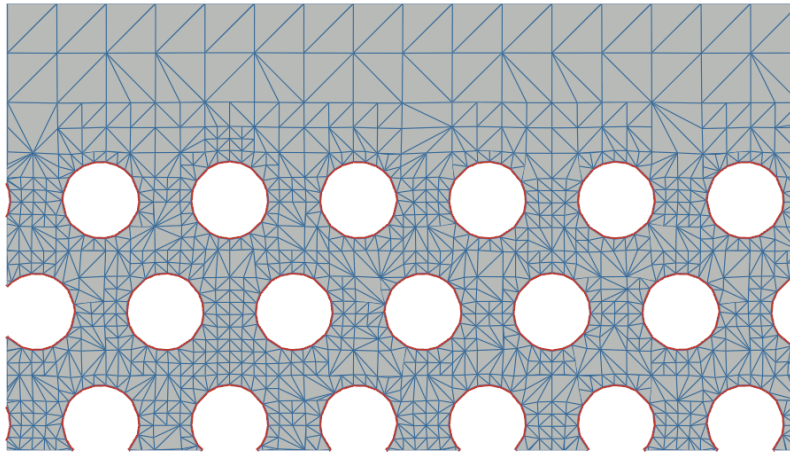
A structured background grid (`blockMesh`, $40 \times 40 \times 21$, 33 600 cells) is refined with `snappyHexMesh`: surface refinement and feature snapping on the tubes, refinement boxes over the chip and bundle, and prism layers. The chip patch is cut from the floor. Body-fitted meshing of all 42 tubes is verified (Fig. 7): snapping yields a complete grid of 299 656 cells passing `checkMesh` (max skewness 1.57, 100% layer addition), confirming the limitation to a fully refined grid is cost, not mesh validity (Table 3).

Refined snapped mesh, mid-plane slice (299,656 cells, checkMesh OK)
body-fitted cells around the tubes; 1 mm surface refinement (cluster: 0.1-0.3 mm)



(a) Mid-plane slice: body-fitted cells with graded refinement and the chip box.

zoom: snapped cells hugging the tube walls



(b) Detail at the tube walls.

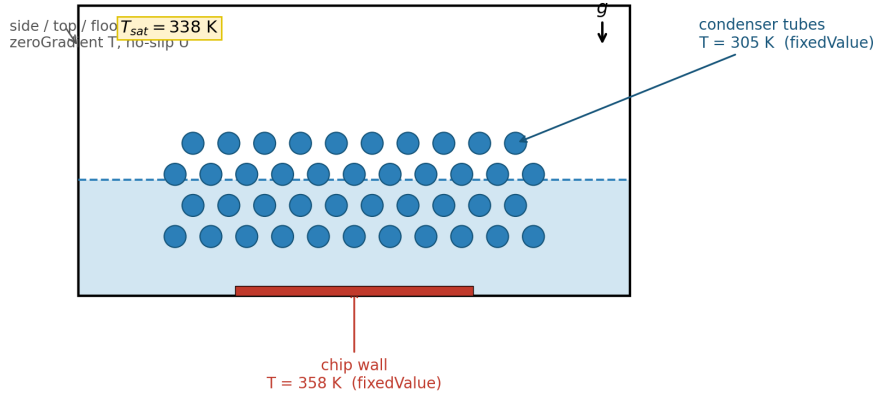
Figure 7: Verified body-fitted snapped mesh (299 656 cells, checkMesh OK).

Table 3: Mesh states.

Mesh	Construction	Cells	Notes
Background	blockMesh	33 600	~ 2 mm
Demonstration	castellation-only	74 044	staircased, robust
Verified snapped	snap, level (1 2)	299 656	checkMesh OK, skew 1.57
Production target	shipped dict	$\sim \text{few} \times 10^6$	0.1 mm to 0.3 mm, snap + layers

5 Boundary, initial, and coolant operating conditions

The chip is a fixed-temperature wall and the condenser tubes a colder fixed-temperature wall; remaining walls are adiabatic no-slip, $T_{sat} = 338$ K, and the pool is initialized below 16.8 mm (Fig. 8, Table 4). These wall temperatures are representative; the calibration matches the simulated boiling curve to the measured one (Section 7).

**Figure 8:** Boundary and initial conditions on the x - z cross-section.**Table 4:** Field boundary conditions.

Patch	T	$\mathbf{U}$	p_{rgh}
chip	fixedValue 358 K	noSlip	fixedFluxPressure
condenser	fixedValue 305 K	noSlip	fixedFluxPressure
floor / walls / top	zeroGradient	noSlip	fixedFluxPressure
Internal $T = 325$ K, $\mathbf{U} = 0$, α from <code>setFields</code> ; $T_{sat} = 338$ K.			

Condenser coolant operating point (measured). The condenser tubes carry single-phase water coolant; its measured operating conditions set the cold-side boundary that the fixed-temperature wall idealizes. Table 5 reports the inlet and outlet temperatures, mass flow, the resulting per-tube velocity and Reynolds number, and the chamber pressure for both condensers across the four coolant set points. The flow is laminar and developing in both bundles ($Re \sim 700$ – 1600); the 42-tube bundle runs at higher velocity (smaller bore) and lower chamber pressure at the warm set

points. The coolant temperature rise $T_{out} - T_{in}$ is the calorimetric basis for the condenser duty $Q = \dot{m}c_p(T_{out} - T_{in})$.

Table 5: Measured condenser coolant conditions (per coolant set point; values are run-averaged). V and Re are per-tube, evaluated at T_{in} .

Bundle, chip	Set	T_{in} (°C)	T_{out} (°C)	ΔT (K)	$\dot{m}$ (g/s)	V (m/s)	Re
33, plain	20	20.0	21.7	1.7	60.0	0.235	735
	30	29.2	31.0	1.8	60.0	0.236	910
	40	39.4	41.0	1.6	60.0	0.237	1116
	50	48.3	49.9	1.6	60.0	0.237	1310
33, micro	20	20.2	23.3	3.1	62.5	0.245	771
	40	40.9	44.4	3.4	62.5	0.247	1197
42, plain	20	20.0	22.1	2.1	38.8	0.610	845
	30	30.0	32.5	2.5	38.8	0.611	1061
	40	40.1	42.2	2.2	38.8	0.614	1298
	50	50.0	52.0	2.0	38.8	0.616	1549
42, micro	20	20.0	23.5	3.5	41.0	0.644	893
	30	29.9	34.5	4.5	38.8	0.611	1060
	40	40.0	44.3	4.3	38.8	0.614	1296
	50	49.9	54.9	5.0	38.8	0.616	1547

Chamber pressure ranges 72 kPa to 85 kPa (33-tube) and 37 kPa to 85 kPa (42-tube), falling at the warm set points as T_{sat} drops.

6 Material properties and numerics

Properties are in Table 6; the phase-change coefficients are the uncalibrated placeholders $C_C = 1.0$, $C_E = 0.1$. Time integration is Euler with an adjustable step ($Co \leq 0.4$); convection uses `linearUpwind` (momentum, energy) and `vanLeer` with interface compression for α ; pressure uses PCG/DIC, α uses MULES, within a PISO loop (Table 7).

Table 6: Material properties.

Property	Water	Vapor	Unit
ρ	980	0.30	kg/m ³
ν	4.4×10^{-7}	4.0×10^{-4}	m ² /s
c_p	4190	2030	J/kg/K
κ	0.66	0.025	W/m/K
h_{fg}		2.35×10^6	J/kg
$\sigma = 0.065$ N/m, $Pr_t = 1.0$, laminar			

Table 7: Run control.

Setting	Value
Δt	1×10^{-6} s
Co, Co_α	≤ 0.4
end (demo)	8×10^{-4} s
α	MULES, $c_\alpha = 1$
p_{rgh}	PCG/DIC
PISO	2 correctors
C_C, C_E	1.0, 0.1 (uncal.)

7 Reference data and the validation target

This section assembles what the CFD must reproduce. **The curves and resistances here are measured data and the validated reduced-order model of the same apparatus, not CFD output.** They define the envelope against which the converged CFD (Section 9) will be judged.

7.1 Measured boiling curves and the reduced-order model

Figure 9 overlays the measured boiling curves (points) with the reduced-order model (lines) for both condensers and both chips across coolant set points. The model tracks the data with a per-configuration RMS error of 3.1 K to 5.2 K in surface temperature. The microchannel surface boils at markedly lower superheat than the plain surface at the same flux, and the 33-tube condenser reaches higher flux at a given surface temperature than the confined 42-tube. These are the trends, and the absolute curves, the 3D simulation is required to recover.

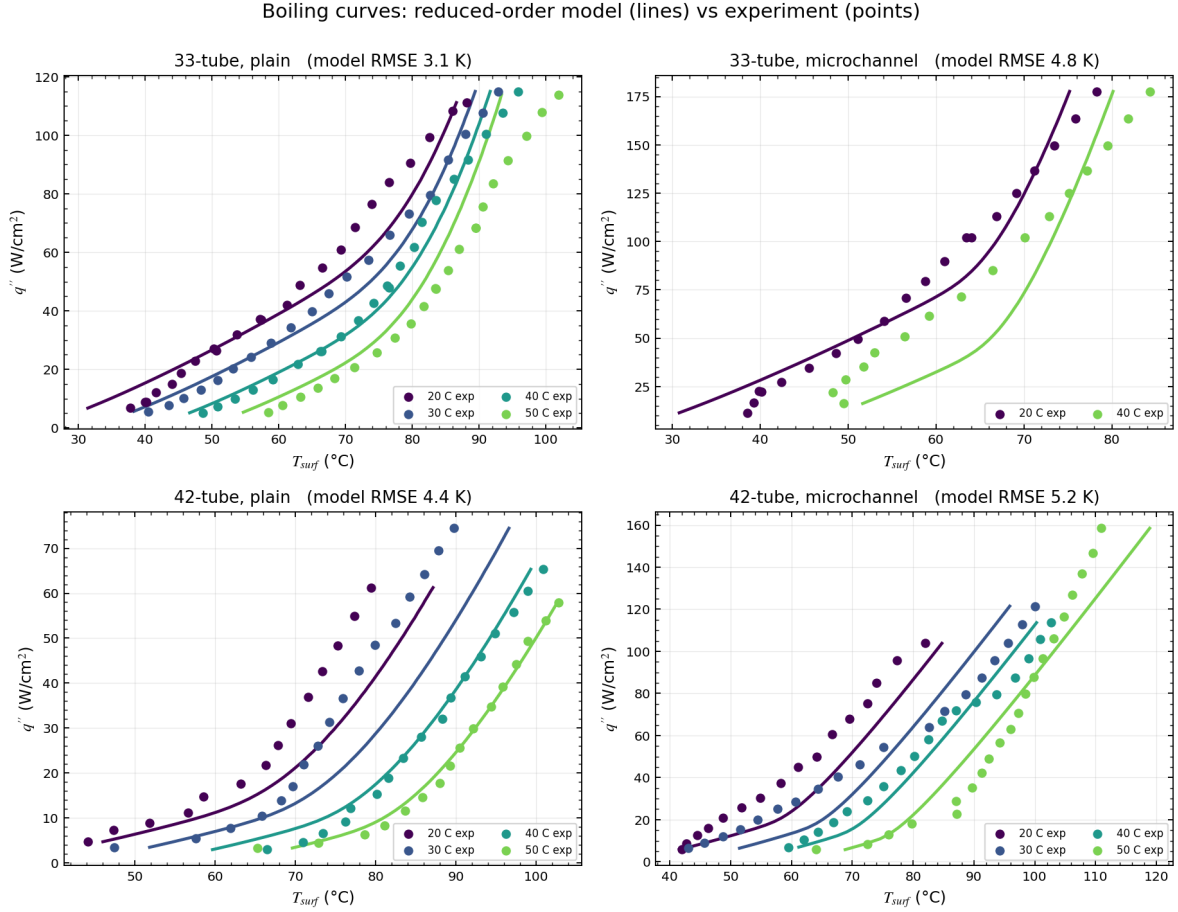


Figure 9: Boiling curves: reduced-order model (lines) vs experiment (points) for both condensers and both chips, colored by coolant inlet temperature. *Reference data and validated reduced-order model; not CFD output.*

7.2 Thermal resistance

The chip-to-coolant path is summarized by the total resistance $R_{tot} = (T_{surf} - T_{in})/Q$, split into a boiling (chip-side) resistance $R_{chip} = (T_{surf} - T_{liq})/Q$ and a pool-to-coolant (condenser-side) resistance $R_{cond} = (T_{liq} - T_{in})/Q$. Figure 10 shows R_{tot} versus heat flux from experiment and from the reduced-order model; Table 8 lists the median breakdown. Two results stand out. First, the condenser-side resistance is far smaller for the open 33-tube bundle (14 mK/W to 15 mK/W) than for the confined 42-tube bundle (41 mK/W to 74 mK/W), consistent with the open bundle being coolant-side limited and the confined bundle carrying a layout penalty. Second, the microchannel

surface roughly halves the chip-side resistance relative to plain (e.g. 17 vs 64 mK/W on the 33-tube), which is the resistance signature of its lower boiling superheat. The reduced-order model reproduces R_{tot} to within a few mK/W except for the 33-tube microchannel, where it over-predicts (44 vs 33 mK/W), the same configuration that carries the largest boiling-curve error.

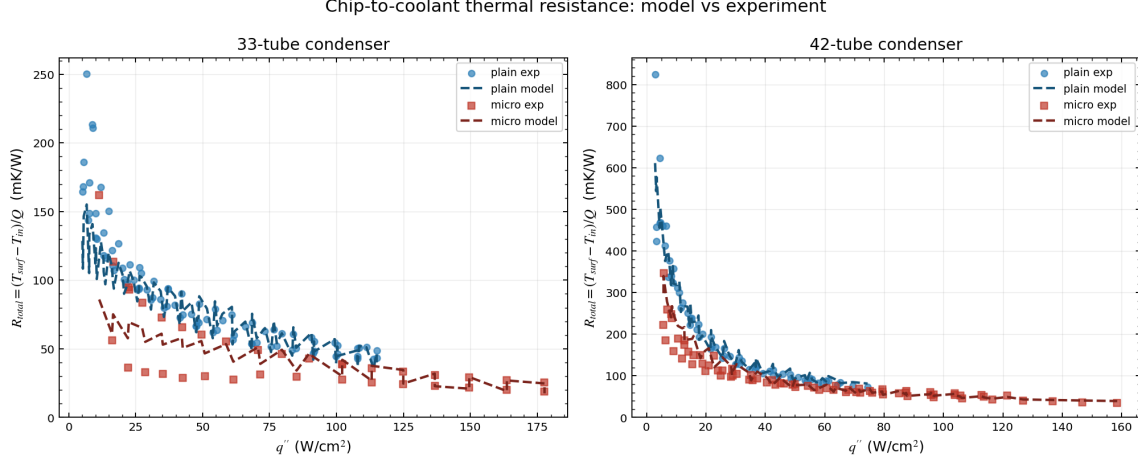


Figure 10: Total chip-to-coolant resistance $R_{tot} = (T_{surf} - T_{in})/Q$ vs heat flux: experiment (markers) and reduced-order model (dashed), per condenser. *Not CFD output.*

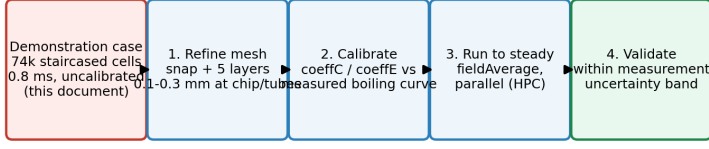
Table 8: Median thermal resistance breakdown (mK/W). R_{tot} is shown for experiment and the reduced-order model; the experimental split into R_{chip} and R_{cond} is from $(T_{surf} - T_{liq})$ and $(T_{liq} - T_{in})$.

Configuration	R_{tot} exp	R_{tot} model	R_{chip} exp	R_{cond} exp
33-tube, plain	78.9	80.5	64.4	13.7
33-tube, microchannel	33.1	44.1	17.3	15.1
42-tube, plain	148.2	161.9	75.1	74.0
42-tube, microchannel	77.1	78.7	31.9	41.1

7.3 Calibration workflow and CHF

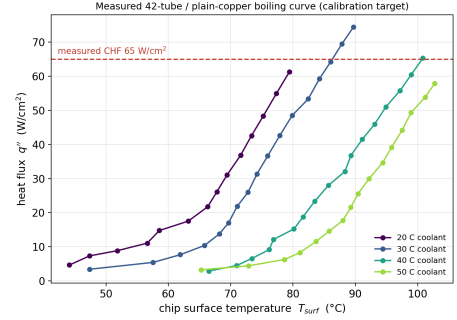
The constant-coefficient closure carries two coefficients that must be calibrated against data (Fig. 11). Calibration sweeps the chip wall temperature for each candidate (C_E, C_C), builds the simulated boiling curve from a chip-patch heat-flux probe ($k_{eff} = \alpha k_w + (1 - \alpha)k_v$), and minimizes the RMS misfit to the measured curve plus a CHF turn-over penalty. The 42-tube/plain target is 64 measured points across four coolant temperatures with a measured CHF of 65 W/cm² (Fig. 11).

From demonstration to validated result



HPC campaign (not runnable in a sandbox); validation requires the quantified uncertainty band

(a) Refine → calibrate → run → validate.



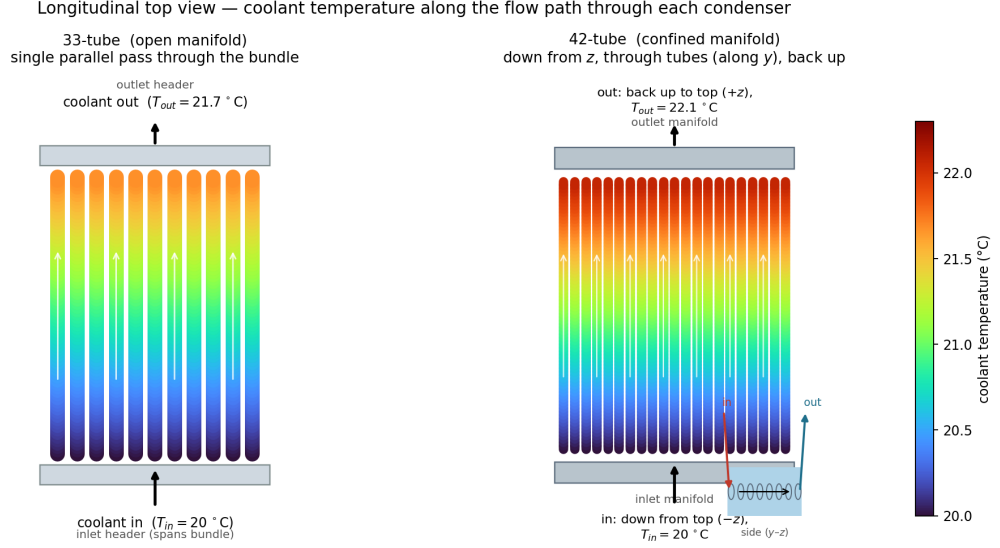
(b) 42-tube/plain target.

Figure 11: Calibration-to-validation workflow and the measured 42-tube/plain calibration target (CHF 65 W/cm²).

8 Condenser coolant-side flow routing and thermal-hydraulics

The two condensers route the single-phase coolant differently, which shapes the cold-side boundary that the fixed-temperature tube walls idealize. The 33-tube bundle uses an open manifold with a single parallel pass: coolant enters a header spanning the bundle, distributes across all tubes, flows along the tube axis, and collects at the outlet header. The 42-tube bundle uses a confined manifold: coolant enters from the top, descends into the inlet manifold, passes through the tubes along their length, and returns up to the top through the outlet manifold (Figs. 12 and 13).

The figures show the coolant temperature along the flow path, computed from a one-dimensional energy balance on the measured condenser duty (Table 5): the coolant warms from the measured inlet to outlet temperature as it absorbs the condenser heat load, shown here at the 20 °C, plain-chip operating point so the two bundles share an inlet. **These are coolant-side thermal-hydraulic estimates on the real geometry, not resolved CFD fields;** in the VOF case the tubes are fixed-temperature walls and the coolant interior is not simulated. The temperature rise is modest in a single pass (1.7 K for the 33-tube, 2.1 K for the 42-tube at this point, the latter warmer at outlet because its flow is lower, Table 5) and scales with heat load, reaching about 5 K for the 42-tube microchannel near CHF.

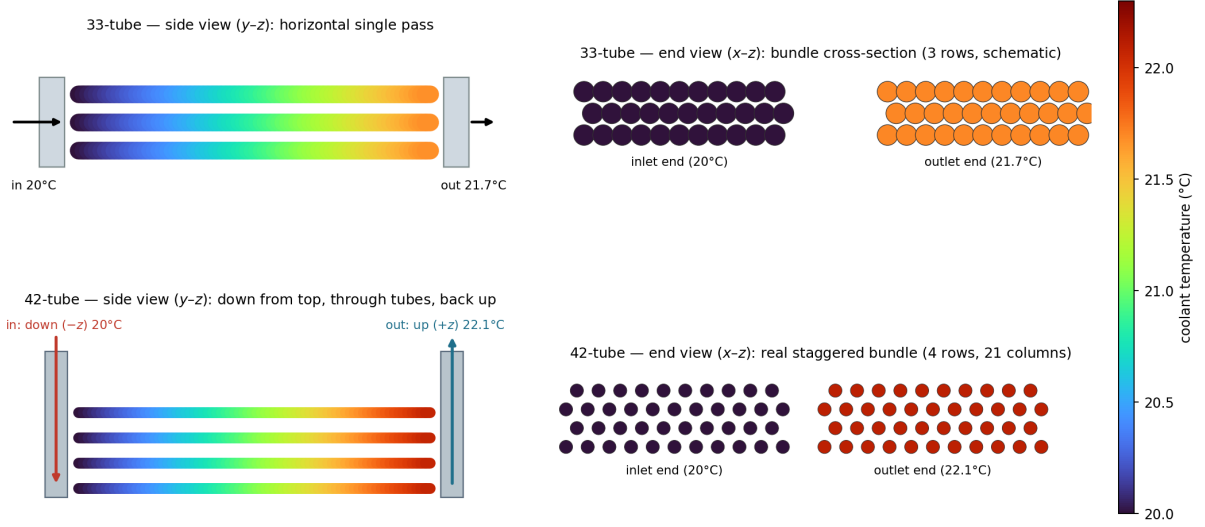


Coolant-side thermal-hydraulic estimate from the measured duty ($Q = \dot{m}c_p\Delta T$, $20\text{ }^{\circ}\text{C}$ set, plain chip); tubes are walls in the VOF case. Not a resolved 3-D CFD field. Routing per described manifold geometry.

Figure 12: Longitudinal top view: coolant temperature along the flow path. The 33-tube (open manifold) makes a single parallel pass; the 42-tube (confined manifold) enters from the top, runs through the tubes, and returns up (side y - z inset). *Coolant-side thermal-hydraulic estimate from the measured duty; not a resolved CFD field.*

The top view (Fig. 12) shows the longitudinal warming along the tube axis. The side view (Fig. 13, left column) contrasts the in-plane single pass of the open 33-tube bundle with the down-through-up routing of the confined 42-tube. The end cross-sections (Fig. 13, right column) show the bundle packing, the real four-row, 21-column staggered 42-tube layout and the schematic three-row 33-tube, with the inlet-to-outlet temperature difference. Resolving the coolant-side velocity, pressure drop, and header maldistribution would require a separate internal-flow or conjugate simulation (Section 11).

Condenser coolant flow — side and end views (coolant temperature along the path)



Coolant-side thermal-hydraulic estimate from the measured duty (20 °C set, plain chip); tubes are walls in the VOF case. Not a resolved 3-D CFD field. Routing per described manifold geometry.

Figure 13: Side (y - z) and end (x - z) views of the coolant path. Left: the 33-tube horizontal single pass and the 42-tube down-from-top / through-tubes / back-up routing. Right: bundle cross-sections at the inlet and outlet ends, with the real staggered 42-tube layout. *Coolant-side thermal-hydraulic estimate; not a resolved CFD field.*

9 Post-processing, diagnostics, and field contours

The case is analyzed with a PyVista post-processor that reads the OpenFOAM time directories through the same reader ParaView uses (Fig. 14). Two data streams are processed: the instantaneous fields, for interface dynamics and snapshots, and the `fieldAverage` (time-averaged) fields, for the stationary quantities that are compared with data. Geometry and the interface are drawn from the mesh patches and the $\alpha = 0.5$ isosurface.

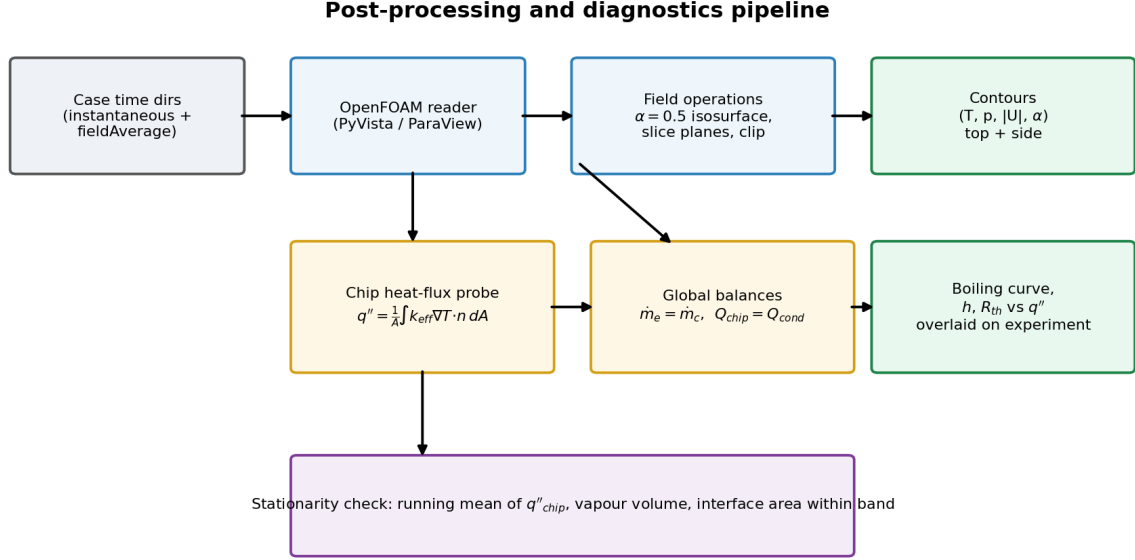


Figure 14: Post-processing and diagnostics pipeline, from the case time directories to contours, global balances, and the boiling curve / resistance overlaid on experiment.

9.1 Chip heat-flux probe and the simulated boiling curve

The simulated heat flux is the phase-weighted conductive flux integrated over the chip patch and normalized by its area,

$$q''_{chip} = \frac{1}{A_{chip}} \int_{A_{chip}} k_{eff} \nabla T \cdot \mathbf{n} dA, \quad k_{eff} = \alpha k_l + (1 - \alpha) k_v, \quad (8)$$

evaluated on the time-averaged field. Sweeping the chip wall temperature and recording the stationary q''_{chip} traces out the simulated boiling curve, which is the quantity fitted to the measured curve during calibration and overlaid on Fig. 9 for validation. The boiling heat-transfer coefficient and the simulated thermal resistance follow directly,

$$h = \frac{q''_{chip}}{T_{surf} - T_{sat}}, \quad R_{tot}^{CFD} = \frac{T_{surf} - T_{in}}{q''_{chip} A_{chip}}, \quad (9)$$

the latter compared against the measured and reduced-order values of Table 8.

9.2 Global balances and verification

At statistical stationarity the chamber must close two balances (Fig. 15), which serve as solver-level verification independent of any experimental comparison: net evaporation equals net condensation, and the chip heat input equals the condenser heat removal, both carried by the latent flux,

$$\int_V \dot{m}_e dV = \int_V \dot{m}_c dV, \quad Q_{chip} = Q_{cond} = h_{fg} \int_V \dot{m}_e dV. \quad (10)$$

The condenser side is cross-checked against the measured coolant calorimetry of Table 5, $Q_{cond} = \dot{m}_{cool} c_p (T_{out} - T_{in})$; a converged run that violates these balances beyond a small tolerance is rejected regardless of how well a single field matches.

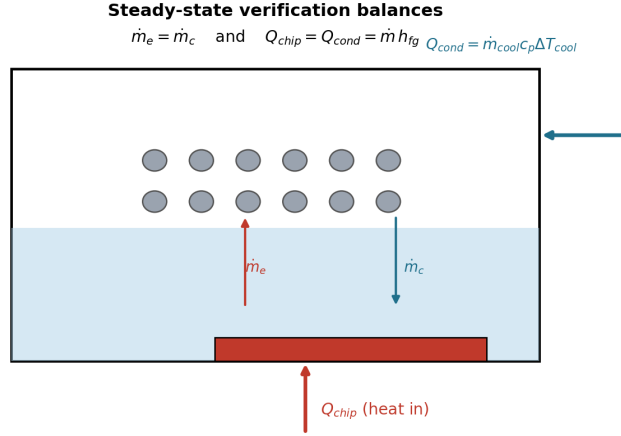


Figure 15: Steady-state verification balances: evaporation at the chip equals condensation at the tubes, and the chip heat input equals the condenser duty set by the measured coolant temperature rise.

9.3 Stationarity, convergence, and mesh sensitivity

A boiling simulation never reaches a steady state pointwise; the diagnostics are declared converged when the running means of three integral monitors, the chip heat flux q''_{chip} , the total vapor volume $\int (1 - \alpha) dV$, and the interfacial area $\int |\nabla \alpha| dV$, settle within a fixed band over a sliding window, with their time averages no longer drifting. Mesh sensitivity is then established by repeating the converged run on the mesh levels of Table 3 and reporting the grid convergence index (GCI) on q''_{chip} and CHF; the production grid is selected where the GCI falls below the measurement-uncertainty band (Section 11, item 1).

9.4 Sampling planes, lines, and derived fields

Scalar fields are extracted on two planes and two lines (Fig. 16): a horizontal x - y plane through the bundle (*top view*) and the vertical x - z mid-plane (*side view*) for contours, a vertical line above the chip center for temperature and velocity profiles through the thermal boundary layer and interface, and a horizontal midline through the bundle for the tube-to-tube field variation. From these, the derived fields reported are the heat-transfer coefficient, the interfacial area density $|\nabla \alpha|$, and the local vapor generation rate $\dot{m}_e$. The plane locations and the color treatment are fixed here so the contour figures populate unchanged once the converged run is available.

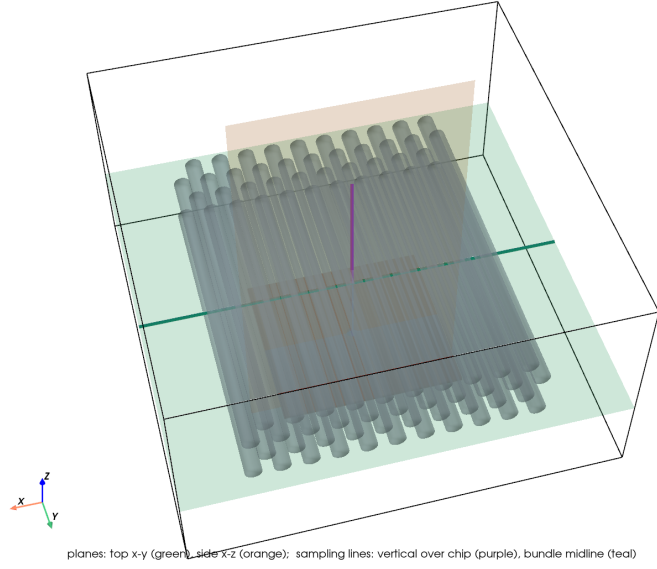


Figure 16: Sampling geometry: top-view (x - y , green) and side-view (x - z , orange) contour planes, with a vertical profile line above the chip (purple) and the bundle midline (teal).

9.5 Verification and single-phase reference fields

Before any phase-change field is trusted, the setup is checked against two fields that are known independently of the VOF solution (Fig. 17). The hydrostatic pressure $p = \rho_l g (z_{fill} - z)$ in the pool is analytic and is the rest-state limit the p_{rgh} solution must recover with $\mathbf{U} \rightarrow 0$. The single-phase conduction field, $\nabla \cdot (k \nabla T) = 0$ with the chip and tubes held at their wall temperatures and the outer walls adiabatic, is solved here on the real x - z geometry and bounds the thermal field in the no-flow, no-boiling limit; it localizes the conductive resistance around the chip and the cold sink at the tubes. **These are verification references, not the VOF phase-change result**, boiling and convection make the actual pool far more uniform and concentrate the gradients into thin interfacial layers. They confirm the boundary-condition setup and provide a conduction-only bound; they do not predict the boiling fields.

Verification / reference fields — NOT the VOF phase-change solution

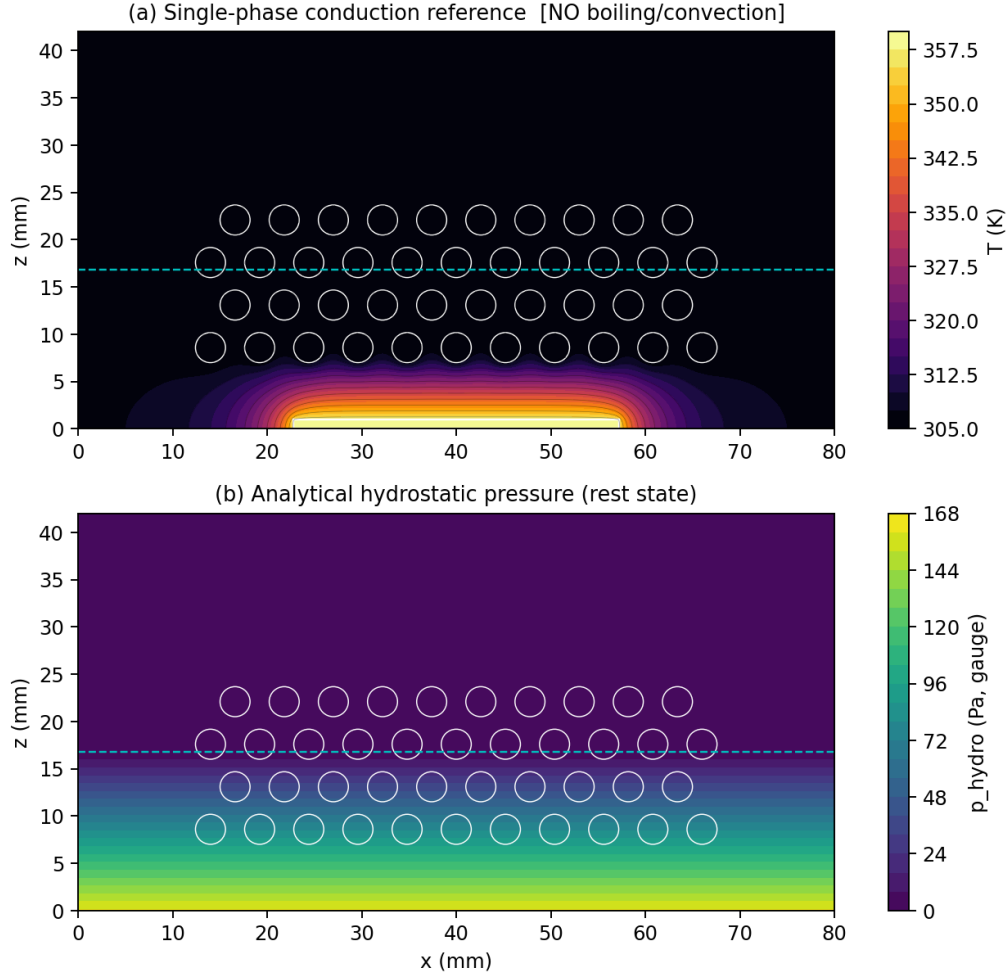


Figure 17: Verification / reference fields on the x - z plane: (a) single-phase steady conduction with the chip and tubes at their wall temperatures (no boiling or convection); (b) analytical hydrostatic pressure. **Neither is the VOF phase-change solution;** they verify the boundary-condition setup and bound the no-flow limit.

9.6 Field contours (pending the converged run)

The following are produced by the production run (Section 7 workflow) and are intentionally left as reserved slots; no surrogate or unconverged fields are shown as results:

- *Temperature contours*, top (x - y) and side (x - z) views, for the 42-tube and 33-tube bundles.
- *Pressure contours*, top and side views, for both bundles.
- *Velocity-magnitude* and interface (α) fields on the same planes, with the profiles along the sampling lines of Fig. 16.
- *CFD-to-experiment overlay*: the simulated boiling curve and thermal resistance from the time-averaged probe, Eq. (8), overlaid on Figs. 9 and 10 for both condensers, with the CHF

turn-over checked against the measured value.

The demonstration temperature and velocity fields carry non-physical extremes and are clamped for display only; they are not presented as results.

10 A reduced-cost two-dimensional demonstration

The three-dimensional case in the preceding sections resolves the real bundle at production mesh resolution and is correspondingly expensive to run to a stationary state. To exercise the same phase-change pipeline at low cost, the project also includes a two-dimensional cross-section case (`openfoam/chamber42_2d`). It is a deliberate simplification with a clearly bounded purpose: exercise the same phase-change pipeline cheaply and establish what such a reduced model can and cannot deliver.

The two-dimensional mesh is generated by `blockMesh` alone, one cell thick in the out-of-plane direction with empty front and back patches. The chip is a native floor patch, obtained by splitting the streamwise direction into three blocks (left floor, chip, right floor) rather than by feature extraction, so no `snappyHexMesh`, surface mesh, or patch surgery is required and the mesh is fully reproducible. The condenser is idealised as the cold top surface at the bundle-top height, held at the coolant temperature; resolving the forty-two circular tubes in two dimensions would require `snappyHexMesh`, whose isotropic refinement subdivides the single out-of-plane cell and breaks the empty-patch two-dimensional constraint. The resolved tube bundle therefore remains the three-dimensional domain, and the two-dimensional case carries the condenser only as a cold boundary. The operating point is set to the measured 42-tube, plain-chip, 20 °C-coolant condition, with saturation 322 K and the cold surface at 293 K.

The intended calibration method is the chip heat-flux probe of Section 9.1 driving a Nelder–Mead search over the evaporation and condensation rate coefficients: the chip wall temperature is swept across the measured range and the time-averaged chip flux is matched to the measured curve, with the evaporation coefficient the primary boiling-branch knob. Running this case on a single workstation core under ESI OpenFOAM v2512 establishes three things about that method, each of which bears directly on the production run.

First, the pipeline is sound end to end. The `blockMesh` mesh passes `checkMesh`, the solver reads the phase-change model and advances stably with the temperature field bounded between the coolant and chip values, and the coded probe compiles at run time and records the chip heat flux. The machinery built for the three-dimensional case runs unmodified on the two-dimensional surrogate.

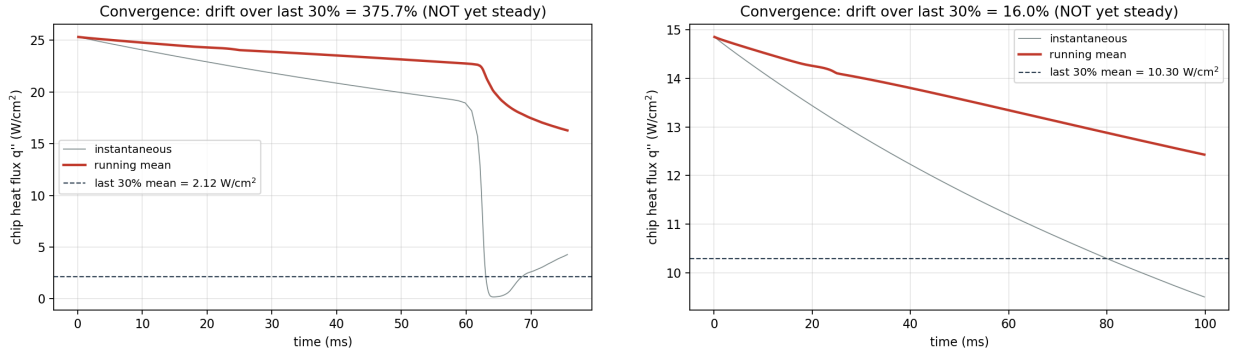
Second, driven beyond the measured critical heat flux the model reproduces the boiling crisis. With the chip held 36 K above saturation, past the measured 65 W cm^{-2} CHF, vapour accumulates at the surface, coalesces into a film after roughly 60 ms, and the chip heat flux collapses toward zero as the low-conductivity vapour blankets the wall. This dryout is the one regime in which the phase change visibly governs the surface flux.

Third, and decisively for the calibration strategy, on the nucleate branch the chip flux is conduction-dominated and essentially independent of the Lee coefficient at this mesh resolution. Holding the wall at 65 °C (16 K superheat) and tripling the evaporation coefficient from 0.1 to 0.3 leaves the chip heat flux unchanged to four significant figures and the domain vapour fraction unchanged in the sixth decimal place (Table 9). The subcooled pool and the absence of resolved

nucleation sites mean that only a thin near-wall band of cells ever exceeds saturation, so the integrated evaporation source is negligible and the surface flux is set by transient conduction into the slowly warming pool rather than by boiling. That bulk warm-up is a seconds-scale diffusion process across the 24mm gap, so at the sub-second times affordable on a single core the chip flux has not stationarised, and no value of the rate coefficient changes that.

Table 9: Chip heat flux at a 65 °C wall (16 K superheat), fixed 0.1 s solve. Tripling the Lee evaporation coefficient leaves both the surface flux and the domain vapour inventory essentially unchanged: at this two-dimensional mesh resolution the phase-change source has no leverage on the measured quantity.

Lee evaporation coeff. C_e	Chip flux q'' (W cm ⁻²)	Domain liquid fraction
0.1	10.30	0.7385951
0.3	10.30	0.7385937



(a) Nucleate point (65 °C wall, 16 K superheat): the chip flux decays monotonically and has not stationarised at 0.1 s.

(b) Past CHF (85 °C wall, 36 K superheat): a vapour film forms near 63 ms and the chip flux collapses toward zero (boiling crisis).

Figure 18: Two-dimensional demonstration runs on a single core under ESI OpenFOAM v2512. The nucleate branch is conduction-dominated and slow to settle; driven past the measured CHF the model exhibits vapour-film dryout. Both traces are early-transient and non-converged, consistent with the Lee-coefficient insensitivity of Table 9.

The consequence is that the quantitative calibration cannot be performed on the two-dimensional surrogate (Figure 18): the coefficient the calibration must fit has no leverage on the quantity it is fitted against, because the mesh does not resolve the boiling. This is a statement about resolution, not cost alone. The resolved-bubble, snapped-and-layered three-dimensional mesh described above is not merely a finer, slower computation; it is the resolution at which the Lee source becomes a dominant term in the surface energy balance and the calibration becomes well-posed. The two-dimensional case is therefore a validated demonstration that the pipeline runs and behaves physically at its limits, and direct evidence for where the quantitative work must be done, not a substitute for it.

11 Next steps

1. **Overlay the measurement-uncertainty bands** on the reference boiling-curve and resistance figures (Figs. 9, 10). The propagated bands (heat flux 4 % to 11 %, CHF $\sim \pm 4$ %),

quantified in the companion reduced-order study, define the tolerance against which any CFD-to-experiment agreement is judged.

2. **Refine and run the 42-tube case** at production resolution: snapped/layered mesh at 0.1 mm to 0.3 mm, calibrate C_E/C_C against the 42-tube/plain boiling curve and CHF, and run to a statistically stationary state.
3. **Produce the field contours** (temperature, pressure, velocity, interface) on the top and side planes of Fig. 16 and insert them in Section 9.
4. **Build and run the 33-tube CFD case** (open-manifold geometry) so the cross-condenser comparison, larger/fewer tubes versus smaller/denser, is resolved in 3D and not only in the reduced-order model.
5. **Resolve the coolant side** with a separate internal-flow or conjugate heat-transfer simulation of each manifold and tube bundle, replacing the fixed-temperature wall with a computed tube-wall temperature and capturing coolant velocity, pressure drop, and header maldistribution; the thermal-hydraulic estimate of Section 8 is the one-dimensional limit of this.
6. **Overlay the converged CFD on experiment** for both condensers (Figs. 9, 10) and report the CFD thermal resistance against Table 8; passing requires lying within the uncertainty band point by point.
7. **Add the microchannel surface** to the CFD, either resolved or as a calibrated sub-grid enhancement, and validate against the microchannel boiling curves.
8. **Verify the CHF mechanism in the converged solution**, checking whether the simulation reproduces the heat-flux turnover and vapor-blanket dryout anticipated by the two-dimensional demonstration (Section 10); the experimental burnout-versus-ceiling determination is carried out in the companion reduced-order study.
9. **Extend to the 66-tube bundle and HFE-7000** once those datasets are available, to test cross-geometry and cross-fluid generalization in 3D.

12 Status and outlook

This document establishes a reproducible 3D VOF phase-change framework on the true geometry: a verified body-fitted meshing pipeline (299 656 cells, `checkMesh` OK), a complete physics, geometry, and numerics specification, the measured condenser coolant operating conditions, and the full set of experimental and reduced-order validation targets, boiling curves and thermal resistances for both condensers. What remains is the computational campaign of Section 11: calibration, the run to stationarity, the field contours, and the CFD-to-experiment overlay, together with the independent prerequisite of quantifying the uncertainty band. On completion, the framework yields resolved boiling and condensation fields that complement, and stress-test, the validated reduced-order model of the same apparatus.

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